Fleetcor Cross-Sell Strategy Enhancement
Final Project Report

Group 3_Fleetcor Project

Group Members:

Dhwani Pathak

Naga Sai Harika Mangina

Mathukutty Manoj

Johanna Susan Eapen

Sameer Shaik

Course:

BANL-6430-03_ Database Management for Business Analytics

Instructor:

Professor Dr. Pindaro Demertzoglou

Program:

MS in Business Analytics

FleetCor
The Global Fleet Card Company

Executive Summary

Fleetcor, a global leader in business payment solutions, offers the Universal Card to existing Fuel-Only cardholders as a way to boost account value and customer engagement. However, without a precise method to predict which accounts will perform well post-conversion, the company risks cross-selling to low-value customers and missing out on high-value ones.

The objective of this project was to develop a data-driven solution to optimize Fleetcor's cross-sell strategy. This involved integrating and cleaning multiple datasets, building a predictive model to identify high-performing accounts, and developing a dashboard that empowers stakeholders to make informed decisions. Using R and Power BI, we created an end-to-end pipeline from raw data to actionable business insights.

Problem Statement

Fleetcor's existing cross-sell strategy lacked performance predictability. Accounts were being promoted to Universal Cards without strong data-backed reasoning. As a result:

- Some Cross-Sell accounts underperformed after onboarding
- High-value NonCross-Sell accounts were overlooked

To address this, the project proposed a Swap Logic Framework:

- **Swap-In:** Add high-performing accounts not yet offered the Universal Card
- **Swap-Out:** Remove existing accounts that are underperforming
- **Stay:** Retain well-performing Cross-Sell accounts

The end goal was to enable smarter marketing allocation, minimize risk exposure, and maximize revenue through intelligent account selection.

Data Overview

Fleetcor provided 11 distinct datasets covering both Cross-Sell and Non Cross-Sell customer segments. These datasets were in Excel format and included:

- **Cross-Sell Account Info:** Basic information for opted-in accounts
- **Non Cross-Sell Account Info:** Baseline fuel-only account data
- **Aging Data:** Account aging buckets (e.g., 30/60/90 DPD)
- **Payment History:** Total amount paid, number of payments, due days
- **NSF Payments:** Number of Non-Sufficient Funds incidents
- **Revenue Files:** Fuel, non-fuel, and net fee revenue
- **Spend Data:** Monthly spending patterns
- **DNB Score:** Commercial risk assessment from Dun & Bradstreet
- **Vantage Score:** Consumer credit score proxy
- **Write-Off Data:** Write-off amounts and flags
- **Segment Score:** Behavioral segment classification (1–10+)

All files were linked by fake_acctcode, allowing for consistent joining and analysis.

Data Cleaning & Merging

Data preparation was performed in R using packages like readxl, dplyr, janitor, and tidyr.

Steps Followed:

- Cleaned and standardized column names using clean_names()
- Removed duplicates and filtered out irrelevant columns
- Replaced NA in numeric columns with 0 using replace_na()

- Distinctly marked opt_in = 1 for Cross-Sell, and opt_in = 0 for Non Cross-Sell
- Merged all datasets using left_join() on fake_acctcode, maintaining 1:1 relationship integrity
- Conducted exploratory checks on skewed columns, outliers, and data distribution

Outcome:

- ✓ A master dataset with over 209,000 unique accounts
- ✓ Over 300+ variables available for modeling and dashboarding
- ✓ Exported as: Fleetcor_Final_Model_Output.csv

Predictive Modeling (Logistic Regression in R)

We designed a classification model to predict whether an account is a high performer or not. The model used logistic regression due to its interpretability and suitability for binary classification.

Target Variable:

- ➢ is_high_performer (1 = top tier accounts, based on revenue & payments)

Independent Variables:

- ➢ Credit Limit Increase (CLI)
- ➢ Payment Amount and Count
- ➢ DPD indicators
- ➢ NSF Count
- ➢ Fuel/Non-Fuel/Net Fee Revenue
- ➢ Vantage Score & DNB Score
- ➢ Behavioral Segment Score

Performance Metrics:

- ✓ **Accuracy:** 90.8% – how well predictions matched actual values
- ✓ **Precision**: 61.3% – out of those predicted high, how many were truly high
- ✓ **Recall:** 44.8% – out of all actual high performers, how many we caught

Interpretation:

While accuracy was high, recall could be improved with further feature engineering or alternate models (e.g., Random Forest, XGBoost). However, the logistic model provided strong interpretability.

Swap Logic Framework

The model's prediction (pred_class) and true performance (is_high_performer) were used to derive the swap_flag:

Actual/Predicted	0	1
0	Stay	Swap-Out
1	Swap-In	Stay

Swap Flag Definition:

- **Stay:** No action required
- **Swap-In:** Predicted high performer, not opted in – offer card
- **Swap-Out:** Underperformer with card – consider opt-out

Swap Flag Results:

- ✓ **Stay:** 182,857 accounts
- ✓ **Swap-In:** 24,730 accounts
- ✓ **Swap-Out:** 2,206 accounts

This triage helps optimize conversion strategy while minimizing risk.

Executive Insights & Visual Summary (Power BI)

An interactive, filterable dashboard was created in Power BI to enable business users to view KPIs, drill into segments, and explore swap logic breakdowns.

The dashboard was not only functional, but also designed to surface strategic executive insights instantly. Below is a summary of the key insights embedded in each visual page:

Prediction Accuracy Page:

- ❖ **Key Visuals:** Confusion matrix, KPI cards for accuracy/precision/recall, swap breakdown.
- ❖ **Insight:** Over 61% of predicted Class 1 accounts are high performers.
- ❖ **Value:** Validates the logistic regression model and supports cross-sell outreach decisions.

Performance Summary Page:

- ❖ **Key Visuals:** Revenue & payment distribution by segment and class.
- ❖ **Insight:** Segments 3 and 6 deliver strongest revenue performance.
- ❖ **Value:** Suggests which segments to prioritize in future campaigns.

Swap Logic Page:

- ❖ **Key Visuals**: Matrix with is_high_performer vs pred_class, bar chart by swap_flag.
- ❖ **Insight:** Largest opportunity lies in Swap-In bucket.
- ❖ **Value:** Enables leadership to confidently activate top prospects and reduce exposure.

Segment-Based Swap Breakdown Page:

- ❖ **Key Visuals:** Swap flag split by behavioral segment; revenue comparison.
- ❖ **Insight:** High-revenue segments show more swap_in potential, while Swap-Outs cluster around lower-tier segments.
- ❖ **Value:** Informs differentiated strategies per segment tier.

Slicer Integration:

- ❖ Allows filtering by portfolio, segment, CLI, region.
- ❖ Empowers business leaders to conduct drill-down analysis in real time.

Insight Cards & Titles:

- ❖ Each visual includes a bold title and optional textbox highlighting the insight.
- ❖ Example: "**Over 61% of Predicted Class 1 accounts are High Performers**"

DAX Formula (Power BI) & R Code (Data Cleaning & Modeling):

- ✓ High_Performer_Pct_Total = DIVIDE(CALCULATE(COUNTROWS('Fleetcor_Final_Model_Output'), 'Fleetcor_Final_Model_Output'[is_high_performer] = 1), COUNTROWS('Fleetcor_Final_Model_Output'))
- ✓ SwapRate_BySegment = CALCULATE(COUNTROWS('Fleetcor_Final_Model_Output'), 'Fleetcor_Final_Model_Output'[swap_flag] = "swap_in") / COUNTROWS('Fleetcor_Final_Model_Output')

```
library(dplyr)

library(readxl)

library(janitor)

cross_sell <- read_excel("Cross-Sell Acct.xlsx") %>%

 clean_names() %>%

mutate(opt_in = 1)

log_model <- glm(is_high_performer ~ cli_amount + total_payment +
vantage_score, data = model_data, family = "binomial")
```

These measures supported KPI cards and matrix visuals to communicate model outcomes clearly.

Techniques Used:

- ➢ Bar Charts, Stacked Columns, KPI Cards

- ➢ Matrix Table with conditional formatting
- ➢ Tooltips, interactive slicers, and DAX measures

Data Source: Fleetcor_Final_Model_Output.csv

Strategic Recommendation Summary

This dashboard enables the marketing, finance, and analytics teams to:

- ➢ Increase revenue by targeting new high-value Swap-In accounts
- ➢ Mitigate risk through removal of Swap-Out underperformers
- ➢ Visualize portfolio health via dynamic drill-downs
- ➢ Align departments using one single source of truth for performance

The use of slicers, tooltips, stacked visuals, and conditional formatting ensures the dashboard serves both analysts and executives.

Strategic Future Scope

- ❖ Model Enhancements: Experiment with ensemble models like Random Forest or XGBoost for higher recall.
- ❖ Enrich Data: Add external datasets such as fuel price trends, regional economic indicators, or market risk indices.
- ❖ Automate Refreshes: Use Power BI Gateway to automate monthly data refreshes and schedule reporting.
- ❖ Deploy Scoring Engine: Move logistic regression model into production for real-time eligibility scoring.

Model Interpretations

- ✓ Logistic Regression Coefficients: Showed CLI amount, total payment, and Vantage Score as strong positive predictors.
- ✓ Segment Influence: Segment 3 and 6 had statistically significant relationships with high performance probability.

- ✓ NSF and Write-Off: High NSF counts and write-off flags were negatively associated with performance.

These interpretations guided variable selection and provided confidence in model logic to business stakeholders.

Conclusion

The Fleetcor Cross-Sell Strategy Enhancement project achieved its core goal of turning raw customer data into a strategic advantage. Through predictive modeling, swap logic analysis, and Power BI dashboarding, we delivered a business-ready solution that:

- ✓ Increases Universal Card adoption among high-value accounts
- ✓ Reduces credit and engagement risk
- ✓ Equips stakeholders with real-time visual decision tools

This approach not only optimizes marketing ROI but sets a precedent for applying analytics across other product lines. The solution is scalable, interpretable, and ready for production deployment.

Submitted by:

- ✓ **Dhwani Pathak**
- ✓ **Naga Sai Harika Mangina**
- ✓ **Mathukutty Manoj**
- ✓ **Johanna Susan Eapen**
- ✓ **Sameer Shaik**

FLEETCOR®